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SUPERSEDING
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March 30, 1970

FEDERAL SPECIFICATION

TUBE, CAPILLARY, BLOOD SAMPLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers glass capillary tubes for use in the determination of blood hematocrit levels and coagulation time.

1.2 Classification. Capillary tubes covered by this specification shall be of the following types and classes, as specified (see 6.1):

1.2.1 Types.

Type I - Heparinized.
Type II - Plain.

1.2.2 Classes.

Class A - Borosilicate.
Class B - Soda lime.

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, Mo, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, Wa.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6630

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Pharmaceutical Association Publication:

National Formulary (N.F.)

U. S. Pharmacopeial Convention, Inc.

Pharmacopeia of the United States (U.S.P.)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, D.C. 20036)

Uniform Classification Committee, Agent

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. [illegible]

3. [illegible]

3.1.1 Class A. Class A capillary tubes shall be of transparent borosilicate glass

[illegible] Class B. Class B capillary tubes shall be of transparent soda lime glass.

3.2 Design. The capillary tube shall be open, without lip or constriction, at both ends.

3.3 Dimensions. The length of the tube shall be 75 ± 0.2 millimeters (mm.). Wall thickness shall be 0.20 ± 0.02 mm. Outside diameter shall be from 1.07 to 1.24 mm. Bore of the tube shall be uniform and not vary in excess of ± 0.025 mm. in 75 mm.

3.4 Construction. One end of the tube shall be fire-polished to [illegible] with rounded plastic closable caps. The [illegible] surface, which is free of constriction; there shall be no change in the toleranced dimensions of the tube, paragraph 3.3, due to fire polishing.

The raw, unpolished end of the tube shall be free of chips, cracks, and burrs, with a smooth slightly round surface. Tube ends shall be cut at 90 degrees to the tube axis and shall not have sharp, chipped, or jagged ends.

3.5 Color coding. Type I (heparinized) tubes, for the purpose of identification, shall be color coded with a [illegible] blue ring located at or near the fire-polished tip of the tube. Type II (plain) tubes, for the purpose of identification, shall be color coded with a [illegible] blue ring located at or near the fire-polished tip of the tube.

3.6 Heparin (for type I). The heparin compound shall be the ammonium salt isolated from the livers, lungs, intestinal mucosa, or other suitable tissues of domestic mammals used for food by man.

3.6.1 Heparin potency assay. The milligrams [illegible] of the ammonium heparin compound shall be equal to or less [illegible] U.S.P. units as determined by 4.6.1.

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3.6.2 Heparin content. Each type I tube shall contain at least 2.0 U.S.P. units of heparin compound as determined in 4.4.2. Inside surface of tube shall be completely covered with a uniform coat of ammonium heparin.

3.7 Performance. The tube shall draw blood to a level which is not more than 2 centimeters (cm.) from the top when tested as specified in 4.4.3.

3.8 Resistance of tubes to centrifugal force. Resistance of tubes to centrifugal force shall be such that no breakage results when the tubes are tested as specified in 4.4.4.

3.9 Labeling. The label of each package of finished capillary tubes shall identify, by reference to appropriate literature or otherwise, the methods by which the article is intended to be used.

3.10 Workmanship. Workmanship shall be first class throughout. Ends of the tubes shall be perpendicular to the tube axis, tubes shall be dry and free of foreign material at time of delivery and before coating for type I, and shall be free of other defects which may detract from their appearance or may impair their serviceability.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table I. Unit of product for examination purposes shall be the tube.

TABLE I. Sampling for examination.

Characteristic	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	0.65
Minor defects	II	2.5
For dimensional examination	S-1	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. Unit of product for test purposes shall be the tube.

TABLE II. Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Heparin potency assay	3.6.1	4.4.1	S-2	0.40
Heparin content	3.6.2	4.4.2	S-2	0.40
Performance	3.7	4.4.3	S-2	1.5
Resistance to centrifugal force	3.8	4.4.4	S-2	1.5

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4.3 Examination. The tubes shall be examined for defects including but not limited to those listed in table III.

Table III. Classification of defects

Categories	Defects
Critical	None defined
Major:	
101	Tube not free of cracks, chips, and breaks.
102	Inside surface not completely coated (type I only).
103	Tube not free of dirt or foreign matter and clean and dry.
104	Tube not straight.
105	Tube not cut at 90 degrees to the axis of the tube.
Minor:	
201	Tube not free of sharp and jagged edges.
202	Tube not free of bubbles

4.3.1 Dimensional examination. The tubes shall be examined for defects in dimension. Any dimension not within tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements

4.4.1 Heparin potency assay. Determine the potency of heparin in accordance with the U.S.P. of N.F. applicable to heparin at date of inspection.

4.4.2 Heparin content. Heparin content shall be determined by the method of assaying sodium heparin in accordance with the [illegible] of U.S.P. applicable at date of inspection. The results obtained shall represent the heparin content on the inner surfaces of the tubes. No heparin from the outside surface shall enter the test sample.

4.4.3 [illegible] technique for hematocrit determination, and the level to which the blood is drawn up within a 30 second period of time shall be noted.

4.4.4 Resistance to centrifugal force. Tubes shall be filled to capacity with distilled water and suspended in a centrifuge. The centrifuge shall be accelerated gradually to a speed of 15,000 revolutions per minute (r.p.m.). Centrifuge shall run at 15,000 r.p.m. for 6 minutes. The centrifuge shall then be shut off and allowed to stop without the use of a brake.

4.5 Examination of preparation for delivery. Examination shall be made to determine whether the packaging, packing, and marking comply with the requirements in section [illegible]. Defects shall be scored in accordance with table IV. The sample unit shall be one shipping container prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. Sampling shall be in accordance with MIL-STD-105, inspection level S-2, with an AQL of 4.0 percent defective

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TABLE IV. Inspection of preparation for delivery

Examine	Defects
Container	Not as specified.
Content	Not as specified.
Material	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or bulged or distorted containers.
Marking	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Unitization	Not as specified.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or Commercial as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit packaging. One hundred tubes of one type and class only shall be packaged in a plastic or glass vial. The vial shall be closed to prevent accidental opening.

5.1.1.2 Intermediate packaging. Twelve vials, packaged as specified in 5.1.1.1 of one type and class only, shall be intermediately packaged in a paperboard box. The box shall be closed to prevent accidental opening.

5.1.2 Commercial. The tubes shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. Packing shall be level A, or Commercial as specified (see 6.1).

5.2.1 Level A. Twenty-four intermediate packages containing only one type of tube shall be packed in a fiberboard box made from weather-resistant fiberboard with a bursting test strength of not less than 275 pounds per adhesive applied to not less than 75 percent of the surface area of contact between the flaps, or with 3 inch wide waterproof tape applied to the full length of the seams and extending over the ends not less than 3 inches. Alternatively, wirebound, sheathed plywood, or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The barrier material shall be sealed at the edges with waterproof tape or adhesive.

5.2.2 Commercial. The packaged tubes shall be packed in fiberboard boxes to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization. When shipments to Government depots are full car or truckload, the shipping containers shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. Marking shall be as specified in the contract or order (see 6.1).

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Packaging and packing required (see 5.1 and 5.2).

Preparing Activity:

GSA - FSS

Orders for this publication are to be placed with General Services Administration acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain federal stocks and other documents not listed herein.

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